Harsath Kaliappa Gounder Thangavel

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Education

York University — **January 2020 – December 2024**
Bachelor of Science in Computer Science (Honours) — *Toronto, Canada*

- Data Structures
- Algorithm Design
- Data Mining
- Database Systems
- Computer Networking
- Compiler Design
- Computer Architecture
- Operating Systems

Work Experience

Privex — **October 2023 – January 2024**
Software Engineer — *London, United Kingdom*

- Developed an AI customer service agent using RAG techniques on a self-hosted LLM to handle common customer inquiries related to technical issues, products, and services.
- Implemented REST APIs in Flask and integrated with Discord API to serve customers on Privex's Discord server.
- Utilized vector databases and MySQL for training data management, allowing dynamic updates from Discord by Privex staff.

CogniPet — **April 2019 – January 2020**
Software Engineer — *Zürich, Switzerland*

- Trained and deployed a deep learning model using the Xception neural network to recognize 120 breeds of dogs with 93% accuracy.
- Implemented REST APIs in Flask for model inference and integrated them with CogniPet's Android and iOS apps using Amazon Web Services (AWS).

Projects

Jix Interpreter | *C interpreter*

- Developed an interpreter for a dynamically typed programming language with support for basic functional programming.
- Implemented features such as fault-tolerant parsing and error handling, dynamic types, arrays, functions with nested blocks and recursion, built-in functions, conditional statements, loops, and control flow statements.
- Added functional programming support, including first-class functions, higher-order functions, and function pointers.
- Tested the interpreter using automated tests written in Jix and integrated with CI/CD pipelines.

Prism Coverage | *Java code coverage engine*

- Developed a code coverage engine supporting control flow (statement, decision, condition, path) and data-flow (All-Defs, All-DU-Paths, All-P-Uses) coverage criteria for a programming language.
- Implemented coverage for built-in and user-defined types, conditional statements, functions with nested blocks, and loops.
- Generated visual HTML reports with color-coded coverage details and summaries for control-flow and data-flow coverage.

Cap Containers | *C containers library*

- Developed a highly generic, header-only containers library in C.
- Supported sets, maps, vectors, hash-tables (open and closed addressing), queues (circular, dynamic, fixed, and priority), stack, lists, and an arena memory allocator.
- Included thread-safe variants with cross-platform support (Windows, Linux, and macOS).
- Tested using automated tests and CI/CD pipelines, and documented with Doxygen.

Blueth | *C++ networking framework*

- Developed a networking framework for Linux for writing event-driven networking services.
- Implemented event-loop abstraction, caches, HTTP parsers and proxies, thread-pool executors, and codecs.

SOCKS5 Handler | *C++ SOCKS5 library*

- Developed a library implementing RFC 1928 and RFC 1929 for a SOCKS5 client with username/password authentication and remote DNS resolution; achieved 30 stars on GitHub.

Technical Skills

Languages:	Python, C++, C, Java, Go, JavaScript, SQL, Perl, Bash, PHP, HTML/CSS, LaTeX
Technologies:	Linux/Unix, MySQL/PostgreSQL, Django/Flask, JWTs, CI/CD pipelines, Neo4j
Developer Tools:	Git, Docker, LXC/LXD, GDB/LLDB, Amazon Web Services